

Unit 3

Physical Layer and its Design Issues

(7 Hours)

3.1 Introduction, Design Issues and Duties of Physical Layer

3.2 Transmission Media: Guided (Twisted Pair, Coaxial Cable, Optical Fiber), Unguided (Radio Wave, Microwave, Infrared and Satellite)

3.3 Ethernet Cable Standards

3.4 Ethernet, Fast-Ethernet and GigaEthernet

3.5 IEEE 802.11 (Wi-Fi) and IEEE 802.15.1 (Bluetooth) Standards

3.6 Signals

3.7 Transmission Impairment: Attenuation, Distortion and Noise

3.8 Multiplexing: TDM, FDM and WDM

3.9 Switching: Circuit Switching, Message Switching, Packet Switching, Virtual Circuit Switching

3.10 Network Performance: Bandwidth, Throughput, Latency, Bandwidth-Delay Product, Jitter

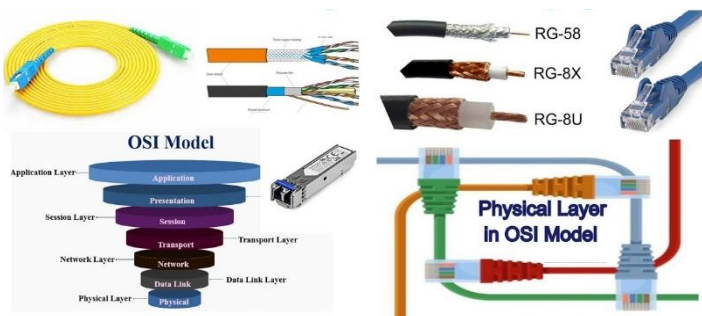
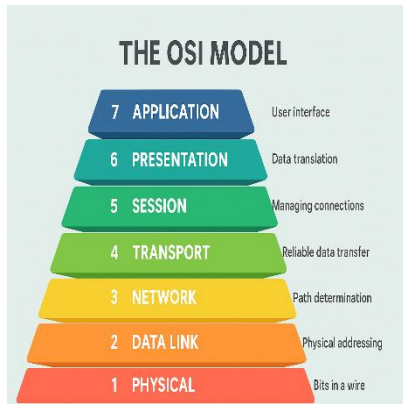
3.1 Introduction, Design Issues and Duties of Physical Layer

The **Physical Layer** is the **first and lowest layer** of the OSI model.

It is responsible for the **actual transmission of raw bits** from one device to another through a physical medium.

This layer deals with:

- cables
- connectors
- voltage levels
- signals
- data rate
- transmission mode
- physical network devices



Simple meaning

If upper layers prepare the data, the physical layer actually **puts the bits on the wire or air**.

Duties of Physical Layer

The main duties of the physical layer are:

1. Transmission of Raw Bits

It sends **0s and 1s** over the communication channel.

2. Representation of Bits

It defines how bits are represented:

- electrical signals
- light pulses
- radio waves

3. Data Rate Control

It specifies how many bits are transmitted per second.

4. Bit Synchronization

It ensures sender and receiver remain synchronized.

5. Line Configuration

It decides whether communication is:

- point-to-point
- multipoint

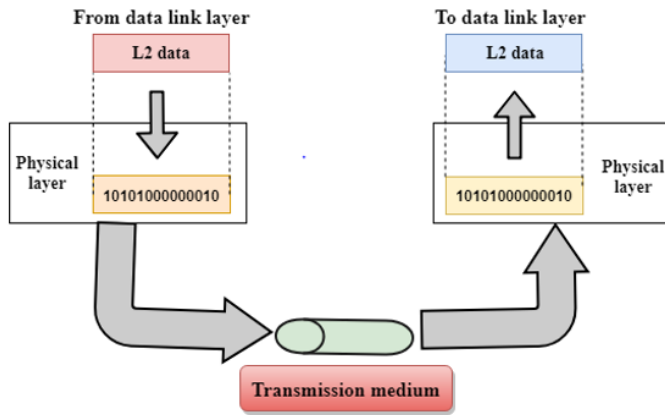
6. Physical Topology

It supports physical arrangement such as:

- bus
- star
- ring

It defines:

- simplex
- half-duplex
- full-duplex



Design Issues of Physical Layer

Important design issues are:

- how bits should be represented
- how sender and receiver synchronize
- what transmission medium should be used
- how much data rate is supported
- how connectors and interfaces are designed
- how signals travel over long distance
- how to reduce interference and loss

3.2 Transmission Media

Introduction

Transmission media means the path or channel through which data travels from the sender to the receiver.

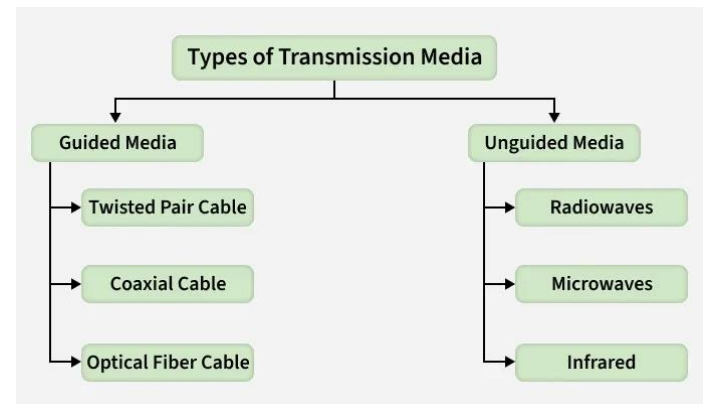


In a communication system, data cannot move by itself. It needs **some medium to carry signals**.

Example:

- when two computers are connected by cable, the **cable is the transmission medium**
- when mobile phones communicate wirelessly, air acts as the transmission medium

Transmission media are mainly divided into **two types**:



1. Guided Media

2. Unguided Media

1. Guided Media

In **guided media**, signals travel through a **physical path**.

The signal is guided inside a cable or wire.

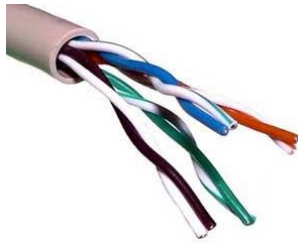
Types of Guided Media:

- Twisted Pair Cable
- Coaxial Cable
- Optical Fiber

1.1 Twisted Pair Cable

A **twisted pair cable** consists of **two insulated copper wires twisted together**.

The twisting helps reduce noise and interference.

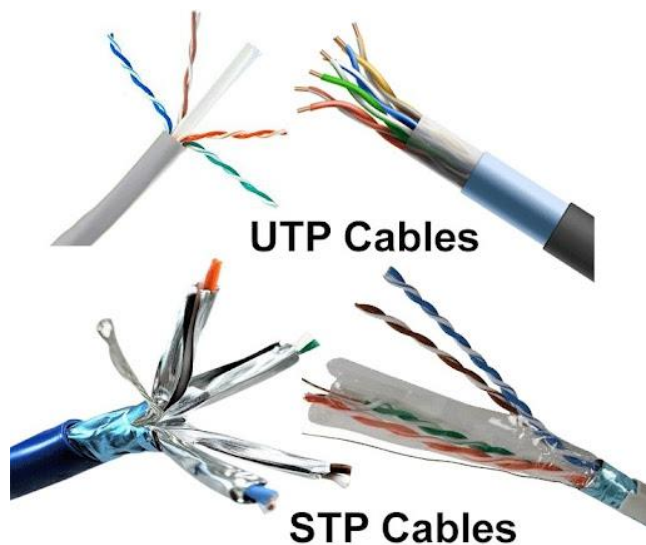


Types:

- UTP (Unshielded Twisted Pair)
- STP (Shielded Twisted Pair)

Uses:

- telephone lines
- LAN networks
- Ethernet connections
- office networking



Advantages:

- low cost

- easy to install
- flexible
- widely used

Disadvantages:

- limited distance
- more affected by noise than fiber
- lower bandwidth than optical fiber

Example:

In a college computer lab, PCs connected to a switch using LAN cable usually use twisted pair cable.

1.2 Coaxial Cable

A **coaxial cable** has:

- one central copper conductor
- insulation layer
- metallic shield
- outer protective cover



It provides better shielding than twisted pair cable.

Uses:

- cable television
- CCTV
- broadband internet
- radio communication



Advantages:

- better protection from interference
- higher bandwidth than twisted pair

- stronger and durable

Disadvantages:

- more expensive than twisted pair
- thicker and less flexible
- installation is harder

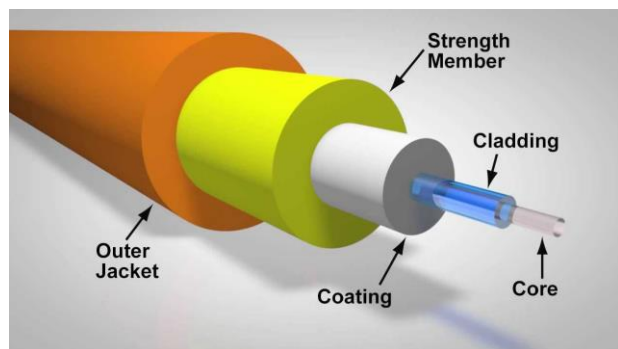
Example:

Cable TV line entering a home is commonly a coaxial cable.

1.3 Optical Fiber

An **optical fiber** is a thin strand of glass or plastic that carries data in the form of **light signals**.

It is the fastest and most secure guided medium.



Types:

- **Single-mode fiber**
- **Multi-mode fiber**

Uses:

- high-speed internet
- backbone networks
- long-distance communication
- telephone networks



Advantages:

- very high bandwidth

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- very high speed

- low attenuation
- not affected by electromagnetic interference
- more secure

Disadvantages:

- expensive
- difficult to install
- fragile compared to copper wire



Example:

Internet service providers use optical fiber to connect cities and major networks.

2. Unguided Media

In **unguided media**, signals travel through **air or free space**.

No physical cable is required.

These are also called **wireless transmission media**.

Types of Unguided Media:

- Radio Wave
- Microwave
- Infrared
- Satellite

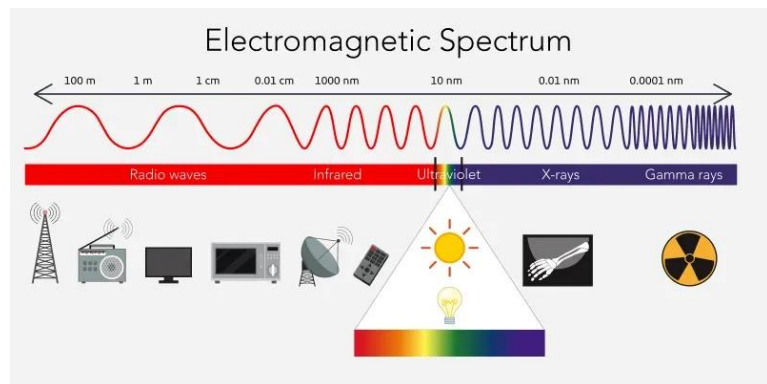
Guided Vs Unguided Transmission Media



2.1 Radio Wave

Radio waves are electromagnetic waves used for wireless communication.

They can travel through air and can pass through walls.



Uses:

- radio broadcasting
- television broadcasting
- Wi-Fi
- mobile communication



Advantages:

- no wires required
- supports mobility
- can cover large areas

Disadvantages:

- interference possible
- lower security
- quality may be affected by environment

Example:

Wi-Fi in a home uses radio waves.

2.2 Microwave

Microwaves are high-frequency radio waves used for long-distance communication.



They usually require **line-of-sight transmission**, meaning there must be a clear path between sender and receiver.

Uses:

- mobile towers
- satellite communication
- long-distance telephone links
- point-to-point communication

Advantages:

- high bandwidth
- good for long-distance communication
- supports large data transfer

Disadvantages:

- affected by weather
- needs line of sight
- installation cost may be high

Example:

Communication between two cell towers often uses microwave links.

2.3 Infrared

Infrared is a short-range wireless medium that uses infrared light waves.

It cannot pass through walls.

Uses:

- TV remote control
- short-range device communication
- some wireless peripherals



Advantages:

- simple
- low cost
- secure for room-level communication

Disadvantages:

- very short range
- cannot pass through obstacles
- not suitable for long distance

Example:

A television remote uses infrared signals.

2.4 Satellite

In **satellite communication**, signals are sent from Earth to a satellite in space and then back to another location on Earth.

It generally uses microwaves.

Uses:

- TV broadcasting
- GPS
- weather monitoring
- internet in remote areas
- military communication



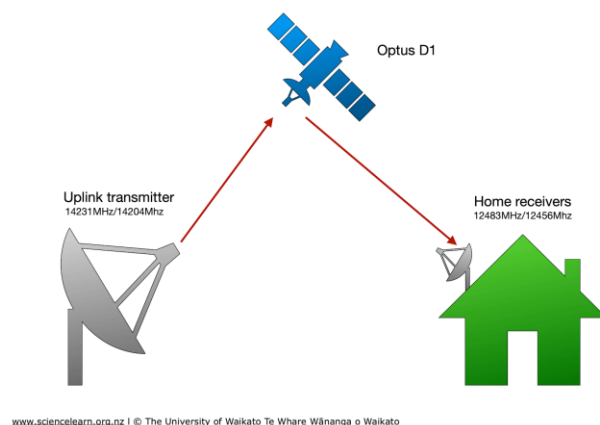
Advantages:

- very wide coverage
- useful in remote and mountainous areas
- supports long-distance communication



Disadvantages:

- expensive
- high latency
- affected by weather in some cases

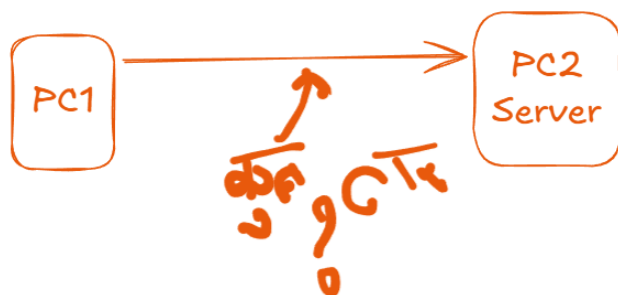


Example:

DTH television and global GPS systems use satellites.

1. Introduction

Ethernet cable standards define the proper arrangement of wires inside an Ethernet cable so that devices can communicate correctly in a network.

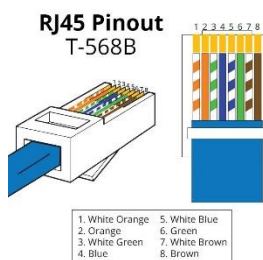


Ethernet cables are commonly used in:

- LAN
- office networking
- school labs
- home routers and switches

The most common Ethernet cable uses an **RJ45 connector** with **8 wires**.

To make the cable correctly, the wires must follow standard color codes.



The two main standards are:

- **T568A**
- **T568B**

1. T568A Standard

In the **T568A** standard, the 8 wires are arranged in this order:

1. **White/Green**

2. **Green**

3. **White/Orange**

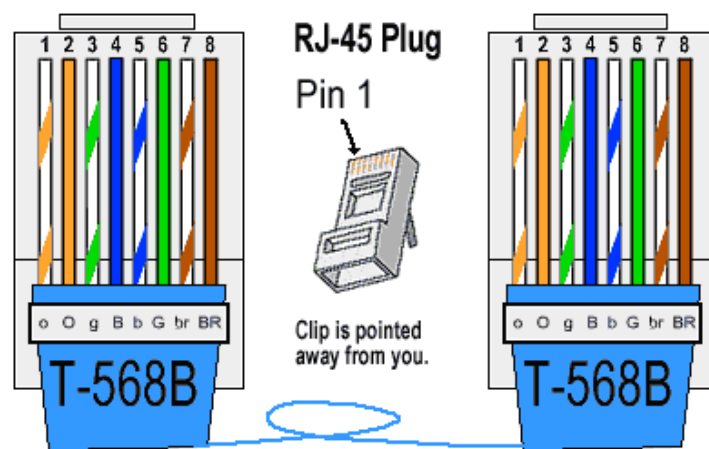
4. **Blue**

5. **White/Blue**

6. **Orange**

7. **White/Brown**

8. **Brown**



2. T568B Standard

In the **T568B** standard, the 8 wires are arranged in this order:

1. **White/Orange**

2. **Orange**

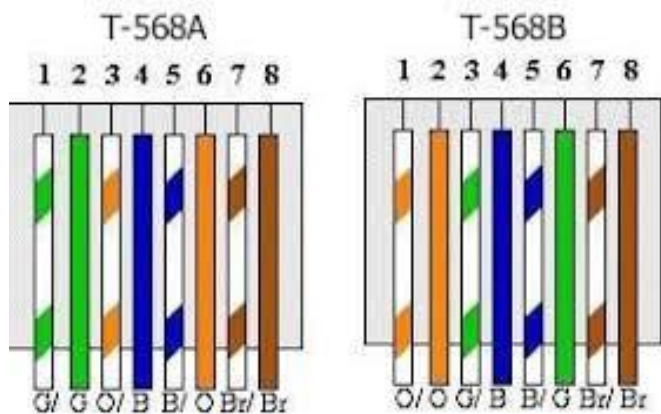
3. **White/Green**

4. **Blue**

5. **White/Blue**

6. **Green**

7. **White/Brown**



3. Why These Standards are Important

Ethernet cable standards are important because they:

- **ensure** correct communication
- reduce wiring **mistakes**
- provide **compatibility**
- help in making **standard** network cables
- make troubleshooting easier

If the wire arrangement is wrong, the cable may not work properly.

4. Types of Ethernet Cables Based on Standards

A. Straight-Through Cable

A **straight-through cable** has the **same standard on both ends**.

Examples:

- T568A to T568A
- T568B to T568B

Uses:

- PC to switch

- PC to hub

- router to switch

Meaning:

The wire order is the same at both ends.

B. Cross-Over Cable

A **cross-over cable** has **different standards on each end**.

Example:

- one end = T568A
- other end = T568B

Uses:

- PC to PC
- switch to switch
- hub to hub
(mainly in older networks)

Meaning:

Transmit and receive pairs are crossed.

5. Ethernet Cable Categories

Ethernet cables also come in different **categories**, which define speed and performance.

Common categories:

- Cat5
- **Cat5e**
- **Cat6**
- Cat6a
- Cat7

- higher category = better speed and performance

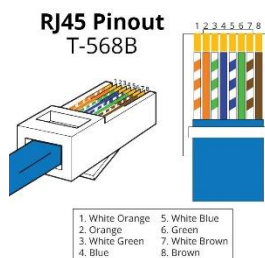
Example:

- **Cat5e** supports up to 1 Gbps
- **Cat6** supports higher speed and better performance

6. RJ45 Connector

The **RJ45 connector** is the connector used at the end of Ethernet cable.

It has **8 pins**, and each pin connects to one wire of the twisted pair cable.

**Use:**

It is plugged into:

- NIC

3.4 Ethernet, Fast-Ethernet and GigaEthernet

Ethernet is one of the most widely used technologies in computer networks, especially in **LAN (Local Area Network)**.

It is used to connect:

- computers
- switches
- routers
- printers
- servers

- switch
- router
- hub

7. Simple Example

Suppose a student wants to connect a computer to a switch in a lab.

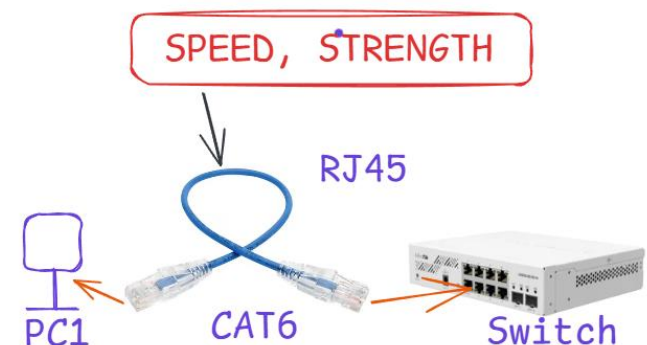
He should usually use a **straight-through cable**, such as:



- T568B on one end
- T568B on the other end

If he wants to connect one old switch directly to another old switch, he may use a **cross-over cable**:

- T568A on one end
- T568B on the other end



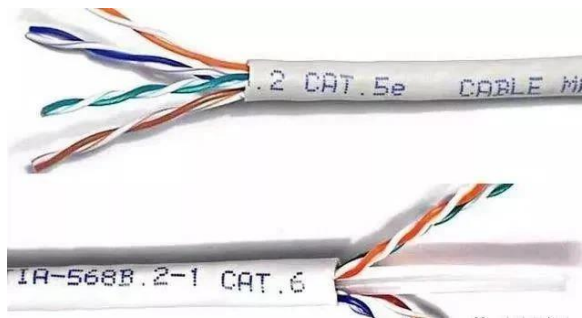
Ethernet defines how data is transmitted in a wired network.

As network demand increased, faster versions of Ethernet were developed:

- **Ethernet**
- **Fast Ethernet**

• Gigabit Ethernet

The main difference among them is **speed**.



1. Ethernet

The original **Ethernet** provides a speed of:

10 Mbps

(10 megabits per second)

It is the basic form of Ethernet used in early LANs.

Features:

- speed = 10 Mbps
- used in small or older networks
- suitable for basic data transfer

Example:

Older office computers connected in a LAN with 10 Mbps speed.

Limitation:

As internet use, file transfer, and multimedia increased, 10 Mbps became too slow.

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2. Fast Ethernet

Fast Ethernet is an improved version of Ethernet.

It provides a speed of:

100 Mbps

(100 megabits per second)

This is **10 times faster** than traditional Ethernet.

Features:

- speed = 100 Mbps
- better for office networks
- supports larger data transfer
- widely used in many LANs

Example:

A school computer lab where many computers share files and internet.

Advantage over Ethernet:

It reduces transfer time and improves performance.

3. Gigabit Ethernet

Gigabit Ethernet is a much faster version of Ethernet.

It provides a speed of:

1000 Mbps = 1 Gbps

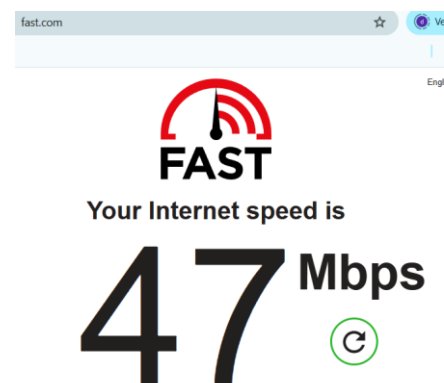
(1 gigabit per second)

This is:

- **100 times faster than Ethernet**
- **10 times faster than Fast Ethernet**

Features:

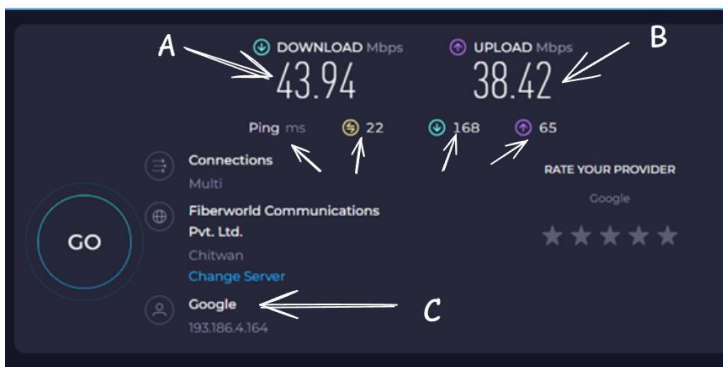
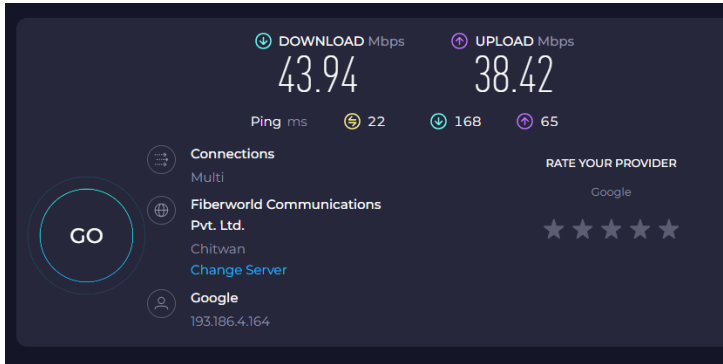
- speed = 1 Gbps



- used in modern LANs
- suitable for high-speed communication
- supports video, server connection, large file transfer

Example:

Modern colleges, data centers, offices, and backbone LAN connections often use Gigabit Ethernet.



4. Simple Speed Comparison

Type	Speed
Ethernet	10 Mbps
Fast Ethernet	100 Mbps
Gigabit Ethernet	1000 Mbps or 1 Gbps

5. Easy Analogy

Imagine three roads for vehicles:

- **Ethernet** = narrow road
- **Fast Ethernet** = wider road
- **Gigabit Ethernet** = highway

The wider the road, the more vehicles can pass quickly.

In the same way:

- higher Ethernet speed means more data can travel in less time

3.5 IEEE 802.11 (Wi-Fi) and IEEE 802.15.1(Bluetooth) Standards

Introduction

IEEE is an international organization that develops standards for networking and communication technologies.



In computer networks, different wireless technologies follow different IEEE standards.

Two important wireless standards are:

- **IEEE 802.11 for Wi-Fi**
- **IEEE 802.15.1 for Bluetooth**

These standards define how wireless devices communicate with each other.

IEEE 802.11 is the standard used for **wireless local area network (WLAN)**, commonly known as **Wi-Fi**.

Wi-Fi allows devices to connect to a network or the internet **without using cables**.

Common devices using Wi-Fi:

- laptop
- smartphone
- tablet
- smart TV
- printer

Main features of Wi-Fi:

- wireless communication
- medium to high range
- high data transfer speed
- used for internet access
- used in homes, offices, schools, colleges, and public places

Uses of Wi-Fi:

- connecting mobile and laptop to internet
- wireless office networking
- smart home devices
- wireless campus network

Example:

When a student connects a laptop to the college router without cable, that is Wi-Fi communication under **IEEE 802.11**.

IEEE 802.15.1 is the standard used for **Bluetooth** communication.

Bluetooth is used for **short-range wireless personal communication**.

It is mainly used to connect nearby devices directly.

Common devices using Bluetooth:

- mobile phone
- wireless headphones
- speaker
- keyboard
- mouse
- smartwatch

Main features of Bluetooth:

- short-range communication
- low power consumption
- used for personal area network (PAN)
- device-to-device connection
- does not need internet for direct pairing

Uses of Bluetooth:

- mobile to headphone connection
- file sharing between two phones
- wireless mouse and keyboard connection
- smartwatch pairing

Example:

When a person connects a Bluetooth earbud to a mobile phone, that communication follows **IEEE 802.15.1** standard.

TYPICAL SYSTEM PARAMETERS OF THE WIRELESS PROTOCOLS				
Standard	Bluetooth	UWB	ZigBee	Wi-Fi
IEEE Spec.	802.15.1	802.15.3	802.15.4	802.11a/b/g
Max data rate (Mbit/s)	0.72	110*	0.25	54
Bit time (μ s)	1.39	0.009	4	0.0185
Max data payload (bytes)	339 (DH5)	2044	102	2312
Max overhead (bytes)	158/8	42	31	58
Coding efficiency ⁺ (%)	94.41	97.94	76.52	97.18
* Unapproved 802.15.3a.		+ Where the data is 10K bytes.		

3. Difference Between Wi-Fi and Bluetooth

Basis	Wi-Fi (IEEE 802.11)	Bluetooth (IEEE 802.15.1)
Full form	Wireless LAN	Wireless Personal Area Network
Range	Larger	Shorter
Speed	Higher	Lower
Power consumption	More	Less
Main use	Internet and network access	Device-to-device connection
Example	Laptop connected to router	Phone connected to headset

4. Simple Analogy

Think of:

- **Wi-Fi** like a **school bus** that carries many students over a longer distance

This means:

- Wi-Fi connects devices over a wider area and carries more data
- Bluetooth is for short-distance personal use

5. Simple Examples

Example of Wi-Fi

In a college:

- one wireless router is installed
- students connect laptops and phones
- all devices access internet without cable

This is **IEEE 802.11 Wi-Fi**

Example of Bluetooth

A teacher connects:

- phone to wireless speaker
- laptop to Bluetooth mouse

This is **IEEE 802.15.1 Bluetooth**

6. Benefits of Wi-Fi

- no cable needed
- easy internet access
- mobility
- can connect many devices
- useful in office, home, campus

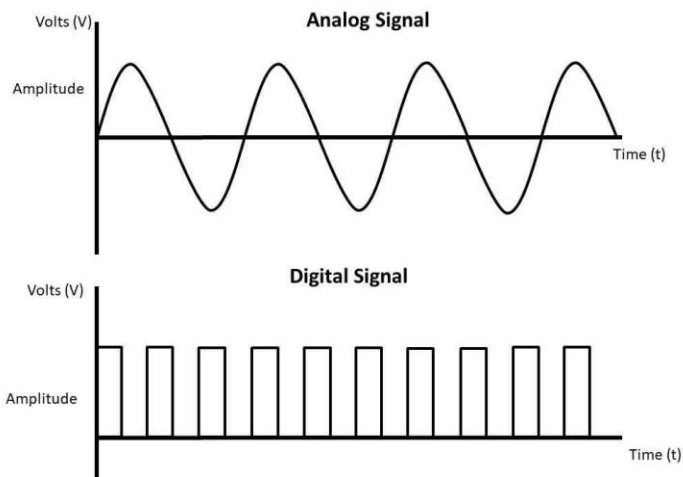
7. Benefits of Bluetooth

- simple short-range connection
- low power use
- easy pairing

- useful for personal devices
- no cable required

3.6 Signals

A **signal** is an electrical, optical, or electromagnetic form of data that carries information from one place to another.



In communication systems, data cannot travel directly. It is first converted into a signal, and then that signal is transmitted through a medium.

Simple meaning

A signal is the **representation of data** used for communication.

Example:

- human voice converted into electrical signal in telephone
- Wi-Fi data sent as radio signal
- optical fiber data sent as light signal

1. Why Signals are Needed

.....

Signals are needed because:

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- data must be transmitted from sender to receiver
- communication devices understand signal form
- information can travel through cable or air only in signal form

Without signals, communication is not possible.

2. Types of Signals

Signals are mainly of **two types**:

1. **Analog Signal**
2. **Digital Signal**

2.1 Analog Signal

An **analog signal** is a **continuous signal** that changes smoothly with time.

It can take infinite values within a range.

Characteristics:

- continuous in time
- continuous in amplitude
- smooth waveform
- more affected by noise

Example:

- human voice

- temperature signal
- radio broadcast
- old telephone signal

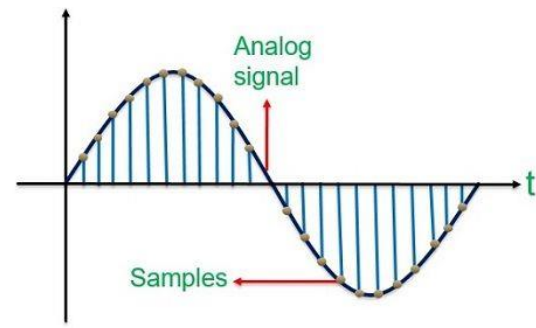
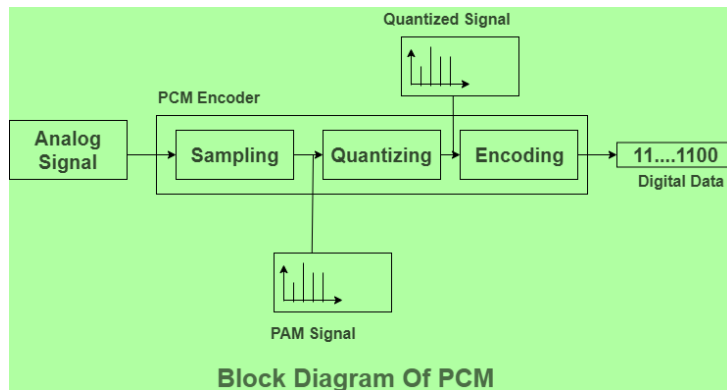
Usually shown as square pulses.

Shape:

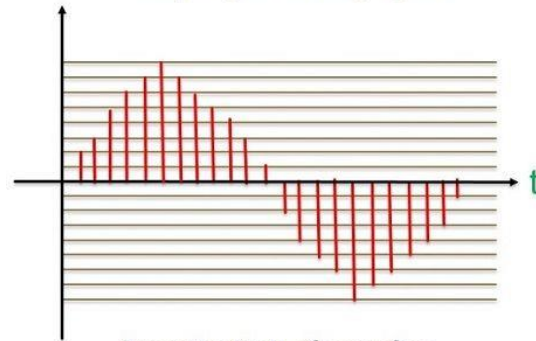
Usually shown as a smooth sine wave.

Simple example:

When a person speaks into a microphone, the sound wave produced is analog.



Sampling of analog signal



Quantization of samples

Simple example:

A computer sending bits 101101 through a network uses digital signal.

2.2 Digital Signal

A **digital signal** is a **discrete signal** that has a limited number of values, usually **0 and 1**.

Characteristics:

- discrete in nature
- represented by binary values
- less affected by noise
- easy to store and process

Example:

- computer data
- digital clock signal
- binary transmission in networks

Shape:

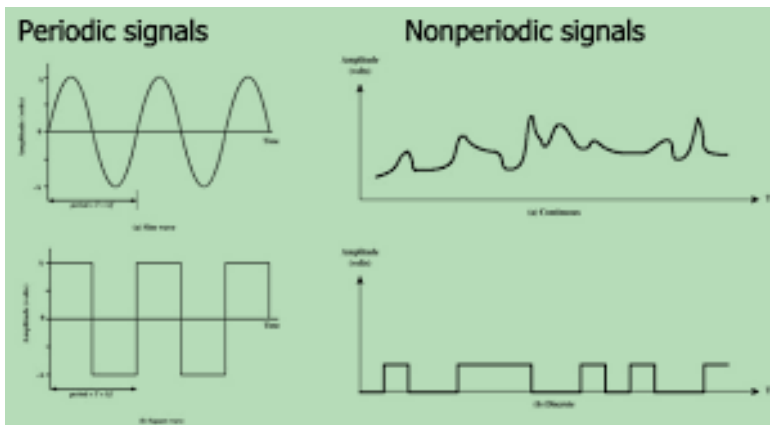
3. Difference Between Analog and Digital Signals

Basis	Analog Signal	Digital Signal
Nature	Continuous	Discrete
Values	Infinite	Limited, usually 0 and 1
Waveform	Smooth	Square pulses
Noise effect	More	Less

Basis	Analog Signal	Digital Signal
Use	Voice, radio	Computers, digital communication

4. Periodic and Non-Periodic Signals

Signals can also be classified as:



4.1 Periodic Signal

A **periodic signal** repeats itself after a fixed interval of time.

Example:

- sine wave
- clock pulse

Meaning:

The same pattern keeps repeating.

4.2 Non-Periodic Signal

A **non-periodic signal** does not repeat its pattern regularly.

Example:

- human speech

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Meaning:

The signal changes irregularly.

5. Important Terms Related to Signals

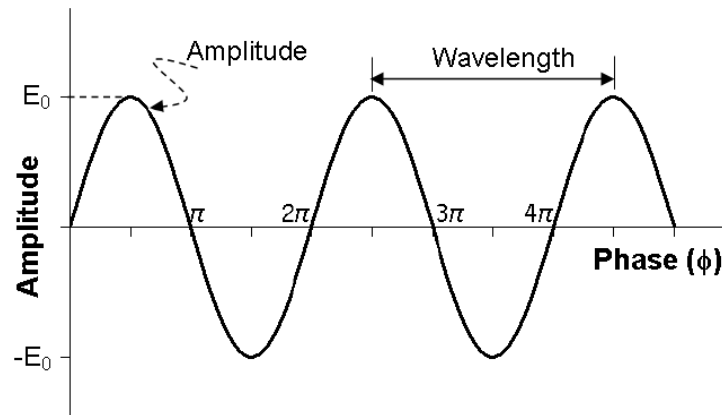
5.1 Amplitude

Amplitude is the maximum height of the signal from the reference level.

It shows the strength of the signal.

Example:

A higher voltage wave has greater amplitude.



5.2 Frequency

Frequency is the number of complete cycles of a signal in one second.

Unit:

Hertz (Hz)

Example:

If a signal completes 50 cycles in one second, its frequency is 50 Hz.

5.3 Time Period

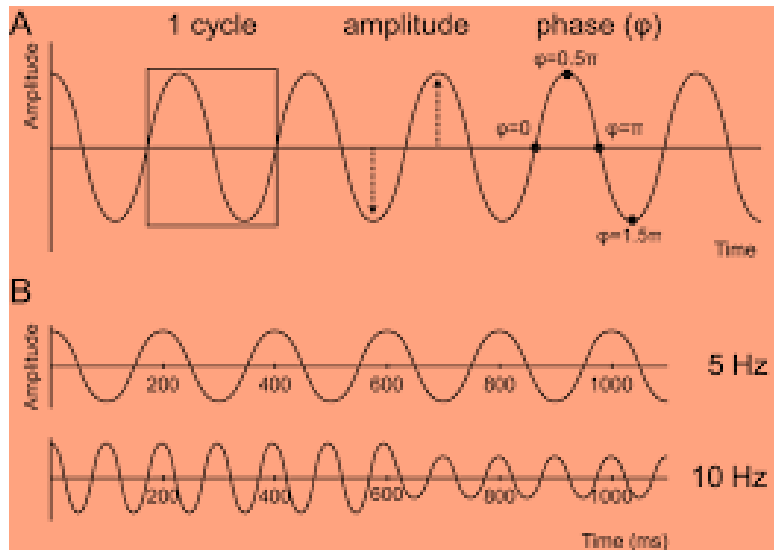
Time period is the time taken to complete one full cycle of a signal.

Formula:

$$T = 1 / f$$

Where:

- T = time period
- f = frequency



5.4 Phase

Phase tells the position of a waveform with respect to time.

It is used when comparing two signals.

6. Simple Example with Data

Suppose a signal completes **100 cycles in 1 second**.

Then,

Frequency = 100 Hz

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If the signal has a maximum height of **5 V**, then:

Amplitude = 5 V

If time for one cycle is:

$$T = 1 / 100 = 0.01 \text{ second}$$

So:

- Amplitude = 5 V
- Frequency = 100 Hz
- Time period = 0.01 sec

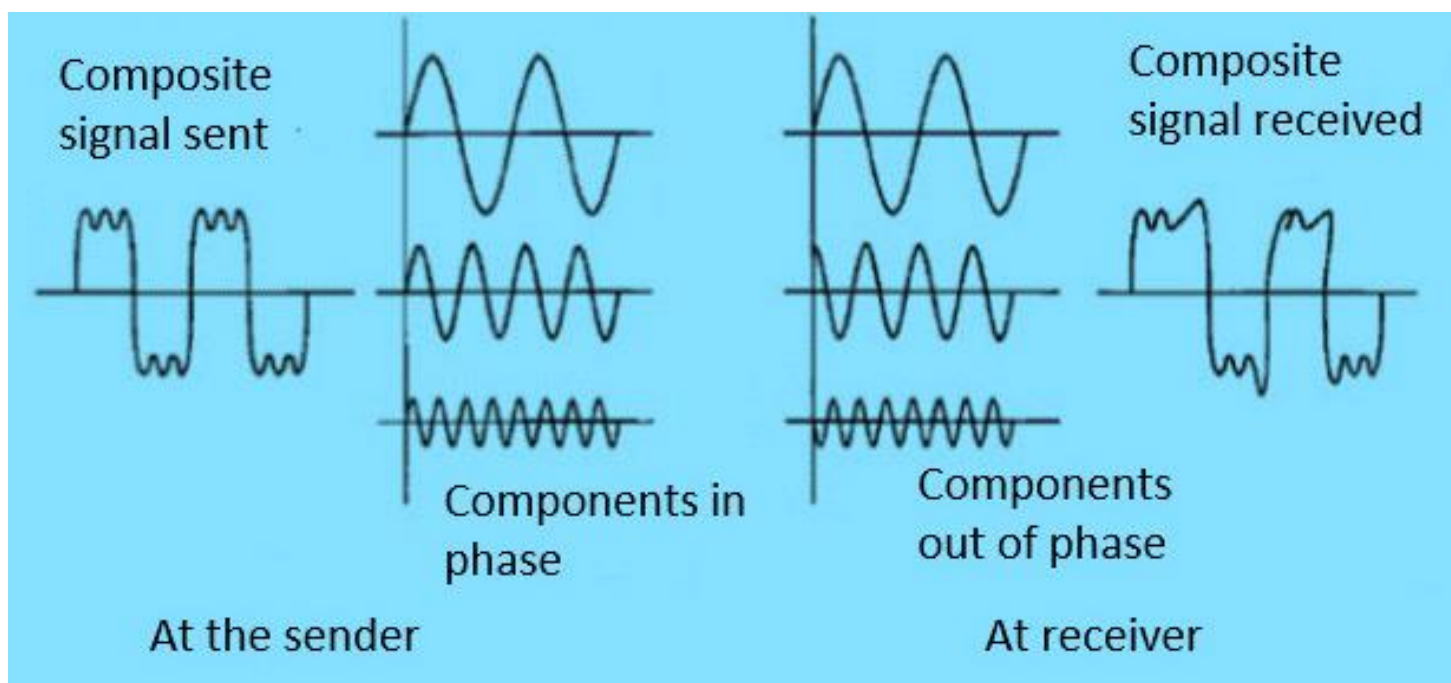
7. Real-Life Examples of Signals

Analog signal examples:

- microphone sound signal
- FM radio signal
- analog thermometer reading

Digital signal examples:

- computer data
- mobile message in binary
- digital watch output
- LAN communication



Introduction

When a signal travels from sender to receiver, it does not always remain perfect.

During transmission, the signal may become weak, changed, or mixed with unwanted signals.

These problems are called **transmission impairments**.

In simple words, transmission impairment means: **any condition that damages the quality of the signal during transmission.**

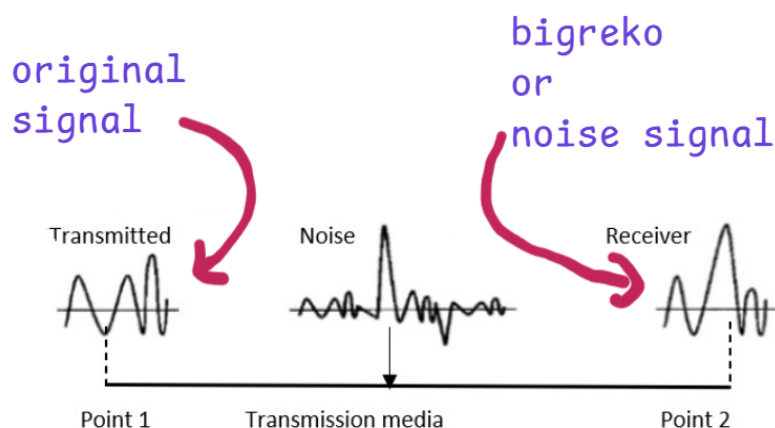
The main types of transmission impairment are:

- **Attenuation**
- **Distortion**
- **Noise**

1. Attenuation

Attenuation means the **loss of signal strength** as the signal travels through the transmission medium.

When a signal moves over a long distance, its power gradually decreases.

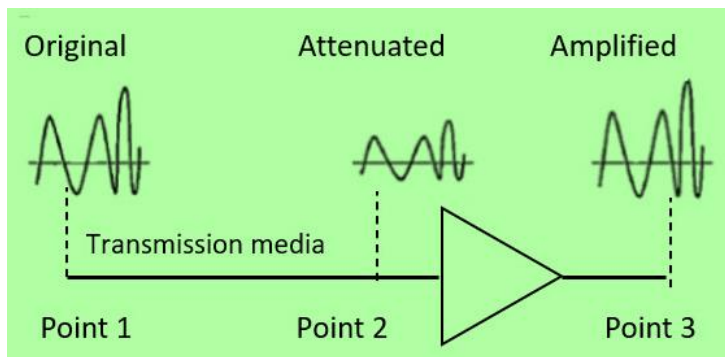


Simple meaning

The signal becomes weaker and weaker with distance.

Example:

If a signal starts with strength **10 V** at the sender and becomes **6 V** at the receiver, then attenuation has occurred.

**Causes:**

- long cable length
- resistance of medium
- energy loss in transmission

Effects:

- weak signal at receiver
- difficult to detect data correctly
- communication errors may occur

Solution:

- use **repeater**
- use **amplifier**
- use better quality medium

Daily life analogy:

Suppose Ram speaks loudly from one side of a field. As the sound travels farther, it becomes weaker. This weakening is like **attenuation**.

2. Distortion

Distortion means a **change in the shape or form of the signal** during transmission.

It happens when different frequency components of a signal travel at different speeds.

So the received signal does not look exactly like the transmitted signal.

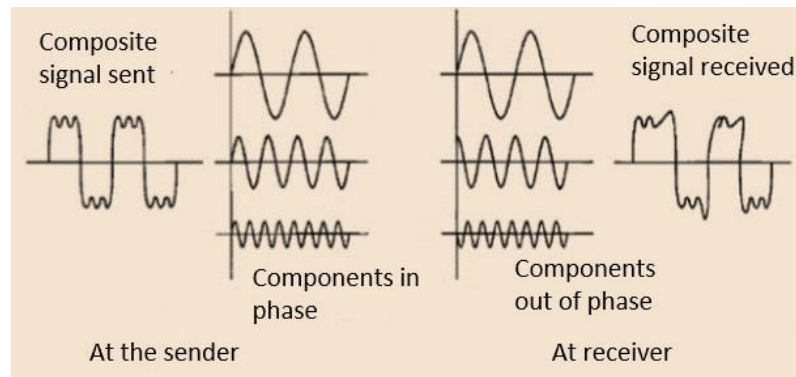
Sanjeev Thapa, Er. DevOps, SRE, CKA, RHCSA, RHCE, RHCSA-Openstack, MTCNA, MTCTCE, UBSRS, HEv6, Research Evangelist

Simple meaning

The signal reaches the receiver, but its shape becomes different.

Example:

If a signal is sent as one waveform but arrives stretched, shifted, or bent, distortion has occurred.

**Causes:**

- different propagation speeds
- medium characteristics
- bandwidth limitations

Effects:

- data may be interpreted incorrectly
- signal quality becomes poor

Daily life analogy:

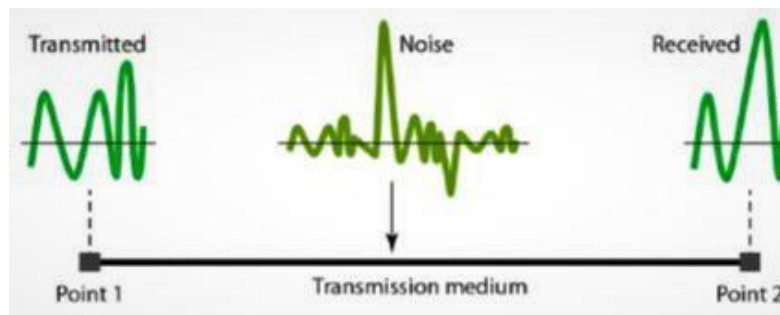
Imagine writing a straight line on paper, but after photocopying badly, it appears curved or stretched. That change in original shape is like **distortion**.

3. Noise

Noise is any **unwanted signal** that gets mixed with the original signal during transmission.

Noise disturbs communication and reduces signal quality.

Noise is extra unwanted disturbance added to the original signal.



Example:

When talking on a phone, if you hear unwanted “shhh” or cracking sound, that is noise.

Causes:

- electrical interference
- thermal effect
- atmospheric disturbances
- nearby signals
- faulty equipment

Effects:

- wrong data may be received
- signal becomes unclear
- communication quality decreases

Types of Noise

1. Thermal Noise

Caused by random movement of electrons in a conductor.

Example:

Present in almost all electronic devices.

2. Induced Noise

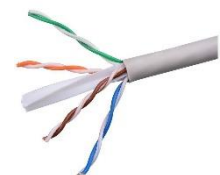
Caused by external sources such as motors, power lines, or electrical devices.

Example:

Noise from nearby electric wire affecting communication cable.

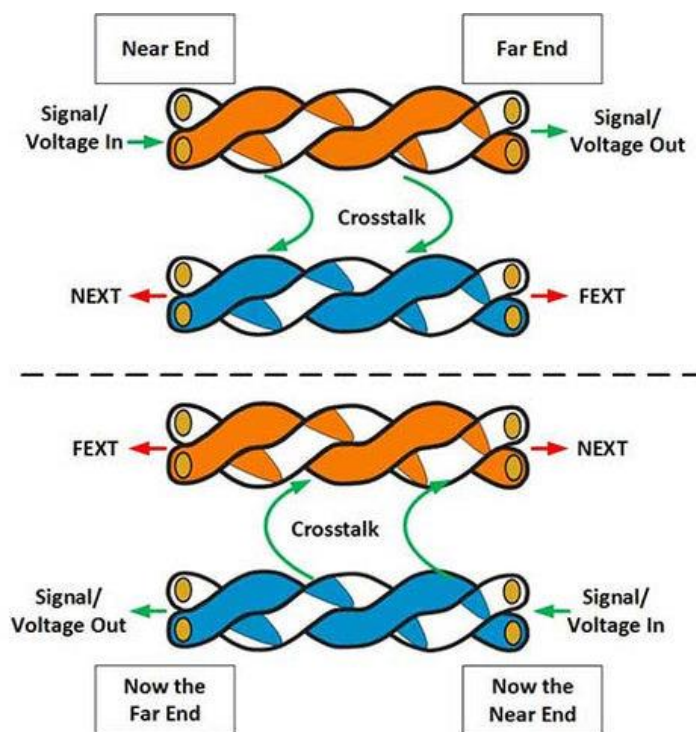
3. Crosstalk

Occurs when one signal interferes with another signal in nearby wires.



Example:

Two telephone lines affecting each other.



4. Impulse Noise

Sudden short-duration noise caused by lightning, switching, or faults.

Example:

A sudden spike in signal during storm.

3.9 Switching: Circuit Switching, Message Switching, Packet Switching, Virtual Circuit Switching

3.10 Network Performance: Bandwidth, Throughput, Latency, Bandwidth-Delay Product, Jitter